Formalizing the Homotopy Extension Property

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Goals of the thesis

- defining the homotopy extension property in Lean
- tools to prove the homotopy extension property
  - retraction criterion
  - homeomorphic space pairs inherit the homotopy extension property from one another
- proving the homotopy extension property for relative CW-complexes

Motivation

Theorem (Cellular Approximation Theorem, [Hat02, Theorem 4.8])

Every map $f\colon X \to Y$ of CW complexes is homotopic to a cellular map. If f is already cellular on a subcomplex $A \subset X$, the homotopy may be taken to be stationary on A.

Theorem (Whitehead's Theorem, [Hat02, Theorem 4.5])

If a map $f\colon X \to Y$ between connected CW complexes induces isomorphisms $f_*\colon \pi_n(X) \to \pi_n(Y)$ for all n, then f is a homotopy equivalence.

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The Homotopy Extension Property

Remark: All maps of topological spaces are assumed to be continuous.

Definition (Homotopy Extension Property (HEP) [Boa16])

A pair of topological spaces (X, A) with $A \subseteq X$ has the *homotopy extension property (HEP)* if for every topological space Y, every map $f \colon X \to Y$, and every homotopy $H \colon A \times [0,1] \to Y$ satisfying $H(a,0) = f(a)$ for all $a \in A$, there exists a homotopy $H' \colon X \times [0,1] \to Y$ such that

$$\begin{aligned} H'(x,0) &= f(x) && \text{for all } x \in X, \\ H'(a,t) &= H(a,t) && \text{for all } (a,t) \in A \times [0,1]. \end{aligned}$$

- (X, X) has the HEP
- $(X, \emptyset)$ has the HEP

Retraction

Let (X, A) be a pair of topological spaces with $A \subseteq X$.

Definition (Retraction)

A map $r \colon X \to A$ is a *retraction* if $r \circ \mathrm{incl}_A = \mathrm{id}_A$.

Equivalently, the function $r \colon X \to A$

- is continuous,
- is fixed on A, i.e. $r(a) = a$ for all $a \in A$.

Retraction Criterion

Lemma (Retraction Criterion)

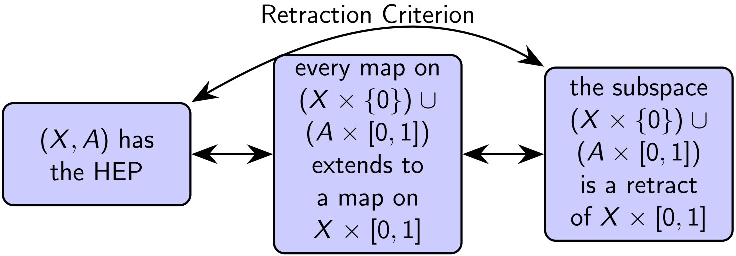
Let X be a topological space, $A \subseteq X$ a subspace. Consider the following two statements:

1. *The pair (X, A) has the HEP.*
2. *The subspace $(X \times \{0\}) \cup (A \times [0,1])$ is a retract of $X \times [0,1]$.*

Then (1) $\implies$ (2). *If, in addition, A is closed in X, the converse* (2) $\implies$ (1) *holds as well.*

Retraction Criterion

Let X be a topological space and $A \subseteq X$ a closed subspace.



HEP for relative CW-complexes

Let X be a CW-complex with skeleta $(X_n)_{n\in\mathbb{N}}$.
Strategy:

1. Any disc relative to its boundary has the *HEP*.
2. The pair of two consecutive skeleta has the *HEP*, i.e. (X_{n+1}, X_n) has the *HEP* for all n.
3. Any relative CW-complex has the *HEP*.

HEP for discs/cubes relative boundary

Theorem ([Boa16])

The pair (D^m, S^{m-1}) has the HEP for all $m \in \mathbb{N}$.

Theorem

The HEP is preserved under homeomorphisms: let (X, A) and (Y, B) be space pairs. If there exists a homeomorphism from X to Y that maps A to B, and (X, A) has the HEP, then so does (Y, B).

Theorem

The pair $([0,1]^m, \partial([0,1]^m))$ has the HEP for all $m \in \mathbb{N}$.

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Definition of the HEP (1/2)

- agreeOn isolates the compatibility condition $H(a,0) = f(a)$
- homotopies have type $X \times \mathbb{R} \to Y$, not $A \times [0,1] \to Y$
- HomotopyExtension : exactly the three conditions of the desired homotopy

```
universe u
variable {X Y : Type u} [TopologicalSpace X] [TopologicalSpace Y] {A :
    Set X}

def agreeOn (f : X → Y) (H : X × ℝ → Y) (A : Set X) : Prop := ∀ (a :
    A), f a = H (a,0)

def HomotopyExtension (H' : X × ℝ → Y) (f : X → Y) (H : X × ℝ → Y)
    (A : Set X) :
   Prop :=
  ContinuousOn H' { p : X × ℝ | p.2 ∈ unitInterval} ∧
  (∀ (x : X), f x = H' (x, 0)) ∧ (∀ (a : A) (t : unitInterval), H (a,t)
     = H' (a, t))
```

Definition of the HEP (2/2)

```
def HEP (X : Type u) [TopologicalSpace X] (A : Set X) : Prop :=
  ∀ (Y : Type u) [TopologicalSpace Y], ∀ (f : X → Y),
  ∀ (H : X × ℝ → Y), Continuous f →
  ContinuousOn H {p : X × ℝ | (p.1 ∈ A) ∧ (p.2 ∈ (unitInterval))} →
  agreeOn f H A → ∃ (H' : X × ℝ → Y), HomotopyExtension H' f H A
```

HEP for a subset instead of a subtype

- motivation: in a CW-complex all skeleta X_n are *subsets* of one ambient space
- HEP_HEP': the two formulations really are equivalent, so nothing is lost

```
open Set.Notation

def HEP' {Y : Type*} [TopologicalSpace Y] (X A : Set Y) : Prop :=
  HEP X (X ↓∩ A)

lemma HEP_HEP' {Y : Type*} [TopologicalSpace Y] (X : Set Y)
    (A : Set (Set.Elem X)) :
    HEP (X.Elem) A ↔ HEP' X (Subtype.val '' A) := by
  unfold HEP'
  rw [Set.preimage_val_image_val_eq_self]
```

Retraction Criterion in Lean

```
structure RetractionOn {X : Type*} [TopologicalSpace X]
    (r : X → X) (B A : Set X) : Prop where
  subset : A ⊆ B
  continuousOn : ContinuousOn r B
  mapsTo : B.MapsTo r A
  fixesOn : ∀ a ∈ A, r a = a

lemma retraction_criterion_closed (hA1 : IsClosed A) :
    HEP X A ↔ ∃ r : X × ℝ → X × ℝ,
    RetractionOn r {p : X × ℝ | p.2 ∈ unitInterval}
    {p : X × ℝ | p.2 = 0 ∨ (p.1 ∈ A) ∧ p.2 ∈ unitInterval}

lemma retraction_criterion_closed' {Y : Type u} [TopologicalSpace Y] (A
     X : Set Y) (hAX : A ⊆ X)
    (hA1 : IsClosed A) : HEP' X A ↔ ∃ r : Y × ℝ → Y × ℝ,
     RetractionOn r
    {p : Y × ℝ | p.1 ∈ X ∧ p.2 ∈ unitInterval}
    {p : Y × ℝ | p.1 ∈ X ∧ p.2 = 0 ∨ (p.1 ∈ A) ∧ p.2 ∈ unitInterval}
```

Transport along a partial homeomorphism

- global homeomorphism $X \simeq Y$ would be stronger
- HEP' is more usfull for partial homeomorphism
- subspaces have to be closed to apply the retraction criterion

```
lemma PartialHomeomorph_HEP' {X Y : Type*} [TopologicalSpace X] [
    TopologicalSpace Y]
    {X1 X2 : Set X} (hX : X2 ⊆ X1) {Y1 Y2 : Set Y} (hY : Y2 ⊆ Y1) (hHEP
      : HEP' X1 X2)
    (f : PartialHomeomorph X Y) (hf_source : f.source = X1) (hf_target :
      f.target = Y1)
    (h2 : f '' X2 = Y2) (hX2closed : IsClosed (X2 : Set X)) (hY2closed :
      IsClosed (Y2 : Set Y)) :
    HEP' Y1 Y2
```

HEP for the disc relative to its boundary

- use `H'`: explicitly constructed earlier
- the primed version follows by `exact`: both statements are definitionally equal

```
lemma HEP_disc_boundary : ∀ (m : ℕ),
    HEP (Metric.closedBall (0 : EuclideanSpace ℝ (Fin m)) 1) (Metric.
     sphere ⟨0, by simp⟩ 1):= by
  intro m Y hY f H hf_cont hH_cont hAgree
  use H' f H
  refine ⟨?_ , ?_ , ?_ ⟩
  · exact H'_ContinuousOn f H hf_cont hH_cont hAgree
  · intro x
    simp [H', aux_fun, mem_closedBall_zero_iff.mp x.prop]
  · exact H'_agreeH f H hAgree

lemma HEP_disc_boundary' : ∀ (m : ℕ),
    HEP' (Metric.closedBall (0 : EuclideanSpace ℝ (Fin m)) 1) (Metric.
     sphere 0 1):= by
  exact HEP_disc_boundary
```

From the disc to the cube

```
theorem exists_homeomorph_image_interior_closure_frontier_eq_unitBall      source
      {E : Type u_1} [NormedAddCommGroup E] [NormedSpace ℝ E] {s : Set E}
      (hc : Convex ℝ s) (hne : (interior s).Nonempty) (hb : Bornology.IsBounded s)
      :
  ∃ (h : E ≃ₜ E),
    ⇑h '' interior s = Metric.ball 0 1 ∧ ⇑h '' closure s =
   Metric.closedBall 0 1 ∧ ⇑h '' frontier s = Metric.sphere 0 1
```

If `s` is a convex bounded set with a nonempty interior in a real normed space, then there is a homeomorphism of the ambient space to itself that sends the interior of `s` to the unit open ball and the closure of `s` to the unit closed ball.

- EuclideanSpace carries the ℓ^2-norm, the plain function type $\mathrm{Fin}\ m \to \mathbb{R}$ the ℓ^∞-norm
- crucially, the above lemma matches closure with closure and frontier with frontier

```
def fun_euclid_max : EuclideanSpace ℝ (Fin m) → (Fin m → ℝ) := fun p ↦ p

def G : (Fin m → ℝ) ≃ₜ (Fin m → ℝ):= (
    exists_homeomorph_image_interior_closure_frontier_eq_unitBall
  convex_euclid_to_max_ball
  nonempty_euclid_to_max_ball bounded_euclid_to_max_ball).choose
```

References I

[Boa16] J. Michael Boardman. *The Homotopy Extension Property*. 2016. URL: https://math.jhu.edu/~jmb/note/hep.pdf (visited on 07/13/2026).

[Hat02] Allen Hatcher. *Algebraic topology*. Cambridge: Cambridge University Press, 2002, pp. xii+544. ISBN: 0-521-79540-0.